

Chapter 6

Momentum and the Conservation of Momentum

1. Linear Impulse and Momentum
2. Impulse-Momentum Theorem and Conservation of Momentum

6-1 Linear Impulse-Momentum Theorem (P167)

1. Impulse-momentum theorem

(The study of time accumulative process about the relationship between the force and movement)

Newton: The change of motion is proportional to the power added, and occurs in the direction of the force along the line.

$$\vec{F}_{\text{ext}} = \frac{d\vec{P}}{dt} = \frac{d}{dt}(m\vec{v}) \quad \vec{P} = m\vec{v} \text{ — momentum}$$

(external forces equal $\vec{P}$ for the first derivative of time)

$$\therefore \vec{F}dt = d\vec{p}$$

(Simultaneous integration of both sides of equality)

$$\therefore \int_{t_i}^{t_f} \vec{F}dt = \int_{p_i}^{p_f} d\vec{p} = \vec{p}_f - \vec{p}_i = \Delta\vec{p}$$

The integral of the force over the time interval during which it acts is called the **impulse**

$$\vec{I} = \int_{t_i}^{t_f} \vec{F}dt$$

$$\vec{I} = \int_{t_i}^{t_f} \vec{F} dt = \Delta \vec{P} = \vec{P}_f - \vec{P}_i \quad \text{impulse-momentum theorem}$$

Net linear impulse delivered to a particle equals the change in linear momentum of the particle.

Descriptions:

(1) $\vec{F}$ should be the sum of all **external forces**. When the force is **constant**, the direction of $\vec{I}$ is as same as the force; if $\vec{F}$ is variable, the direction of $\vec{I}$ is determined by the integral of $\vec{F}$.

(2) Both $\vec{I}$ and $\vec{P}$ are vectors. They have same units and dimensions. This theorem is only hold for Inertial reference frames (IRF).

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(3) In reality, one often use its component form:

$$\vec{I} = I_x \vec{i} + I_y \vec{j} + I_z \vec{k}$$

$$I \begin{cases} I_x = \int_{t_i}^{t_f} F_x dt = mv_{fx} - mv_{ix} \\ I_y = \int_{t_i}^{t_f} F_y dt = mv_{fy} - mv_{iy} \\ I_z = \int_{t_i}^{t_f} F_z dt = mv_{fz} - mv_{iz} \end{cases}$$

Example 6.2(P170)

EXAMPLE 6.2

How Good Are the Bumpers?

GOAL Find an impulse and estimate a force in a collision of a moving object with a stationary object.

PROBLEM In a crash test, a car of mass $1.50 \times 10^3 \text{ kg}$ collides with a wall and rebounds as in Figure 6.4a. The initial and final velocities of the car are $v_i = -15.0 \text{ m/s}$ and $v_f = 2.60 \text{ m/s}$, respectively. If the collision lasts for 0.150 s , find (a) the impulse delivered to the car due to the collision and (b) the size and direction of the average force exerted on the car.

STRATEGY This problem is similar to the previous example, except that the initial and final momenta are both nonzero. Find the momenta and substitute into the impulse–momentum theorem, Equation 6.6, solving for F_{av} .

SOLUTION

(a) Find the impulse delivered to the car.

Calculate the initial and final momenta of the car:

$$\begin{aligned} p_i &= mv_i = (1.50 \times 10^3 \text{ kg})(-15.0 \text{ m/s}) \\ &= -2.25 \times 10^4 \text{ kg} \cdot \text{m/s} \end{aligned}$$

$$\begin{aligned} p_f &= mv_f = (1.50 \times 10^3 \text{ kg})(+2.60 \text{ m/s}) \\ &= +0.390 \times 10^4 \text{ kg} \cdot \text{m/s} \end{aligned}$$

The impulse is just the difference between the final and initial momenta:

$$\begin{aligned} I &= p_f - p_i \\ &= +0.390 \times 10^4 \text{ kg} \cdot \text{m/s} - (-2.25 \times 10^4 \text{ kg} \cdot \text{m/s}) \\ I &= 2.64 \times 10^4 \text{ kg} \cdot \text{m/s} \end{aligned}$$

(b) Find the average force exerted on the car.

Apply Equation 6.6, the impulse–momentum theorem:

$$F_{\text{av}} = \frac{\Delta p}{\Delta t} = \frac{2.64 \times 10^4 \text{ kg} \cdot \text{m/s}}{0.150 \text{ s}} = +1.76 \times 10^5 \text{ N}$$

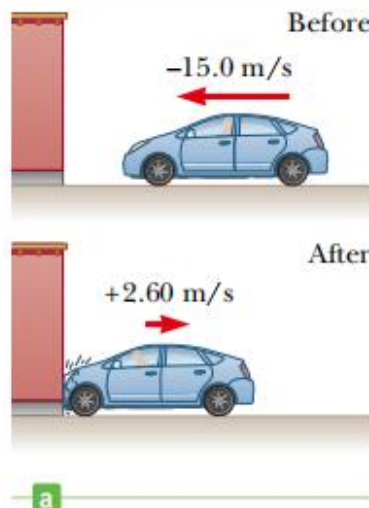


Figure 6.4 (Example 6.2) (a) This car's momentum changes as a result of its collision with the wall. (b) In a crash test (an inelastic collision), much of the car's initial kinetic energy is transformed into the energy it took to damage the vehicle.

6.2 Conservation of Momentum (P172)

1. Momentum theorem of particle system

There are N particles in a system. The equation of motion for particle i is,

$$\vec{F}_{i\text{ext}} + \vec{f}_{i\text{int}} = \frac{d\vec{p}_i}{dt}$$

(*external forces plus internal forces equal P_i for the first derivative of time*)

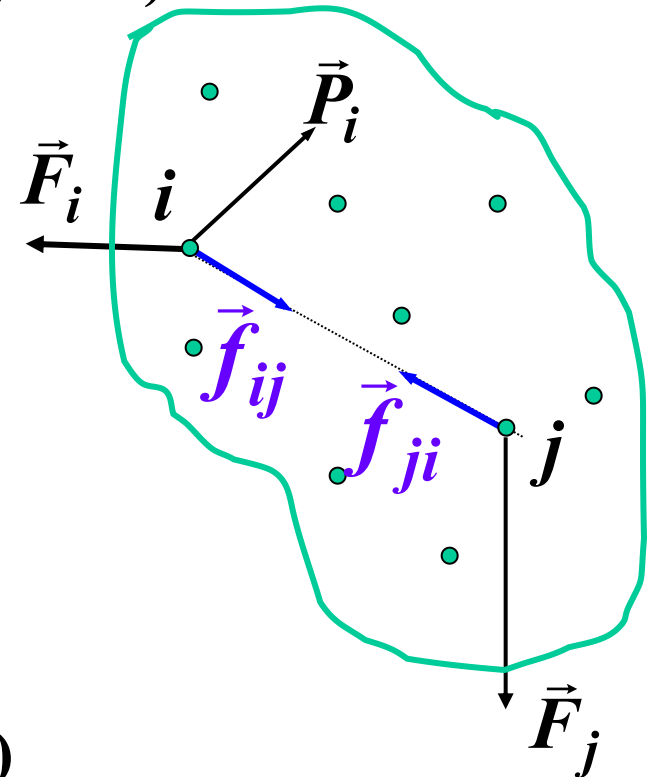
for the whole system is,

(*Summation of both sides of the equation at the same time*)

$$\sum \vec{F}_{i\text{ext}} + \sum \vec{f}_{i\text{int}} = \frac{\sum_{i=1}^N d\vec{p}_i}{dt} = \frac{d\sum_{i=1}^N \vec{p}_i}{dt}$$

where $\vec{F}_{\text{ext}} = \sum_{i=1}^N \vec{F}_{i\text{ext}}$ and $\vec{P} = \sum_{i=1}^N \vec{P}_i$, **note** $\sum \vec{f}_{i\text{int}} = 0$

(**Internal force** is the force and **reaction force**)



$$\vec{J} = \int \vec{F}_{\text{ext}} dt = \Delta \vec{P} \quad \text{— impulse-momentum theorem of particle system}$$

The impulse of external forces acting on a system = the change of momentum of the **entire** system.

If $\vec{F}_{\text{net}} = \mathbf{0}$, then $\vec{P} = \sum_{i=1}^N \vec{p}_i = \vec{C}$ conservation of momentum

If **no net external force** acts on a particles' system, total linear momentum of the system cannot change.

No exceptions to this law have been discovered. So it is considered to be more fundamental than Newton's second law in both **macroscopic** and **microscopic** physical worlds.

If $\vec{F}_{\text{net}}=0$, then

$$\vec{P} = \sum_{i=1}^N \vec{p}_i = \vec{C}$$

conservation of momentum

 **Explanations:**

(1) Conditions of validity: $\sum \vec{F}_{\text{ext}} = 0$, or $\vec{F}_{\text{ext}} \ll \vec{F}_{\text{int}}$

Net forces can be **neglected** in the case of collision, strike, explosion, ... ,and so on.

(2) **Conclusion:** **Only external force** can change the **state of motion** of a system.(Internal force transmits momentum in the system)

(3) If component of net external force on a system along an axis is zero, the component of **momentum of the system along that axis does not change.**

(4) It is **only suitable** for **inertial** reference frame.

Example 6.3(P174)

EXAMPLE 6.3 The Archer

GOAL Calculate recoil velocity using conservation of momentum.

PROBLEM An archer stands at rest on frictionless ice; his total mass including his bow and quiver of arrows is 60.00 kg. (See Fig. 6.9.) (a) If the archer fires a 0.030 0-kg arrow horizontally at 50.0 m/s in the positive x -direction, what is his subsequent velocity across the ice?

SOLUTION

(a) Find the archer's subsequent velocity across the ice.

Write the conservation of momentum equation for the x -direction.

Let m_1 and v_{1f} be the archer's mass and velocity after firing the arrow, respectively, and m_2 and v_{2f} the arrow's mass and velocity. Both velocities are in the x -direction. Substitute $p_i = 0$ and expressions for the final momenta:

Solve for v_{1f} and substitute $m_1 = 59.97$ kg, $m_2 = 0.030$ 0 kg, and $v_{2f} = 50.0$ m/s:

$$p_{ix} = p_{fx}$$

$$0 = m_1 v_{1f} + m_2 v_{2f}$$

$$v_{1f} = -\frac{m_2}{m_1} v_{2f} = -\left(\frac{0.030\ 0\ \text{kg}}{59.97\ \text{kg}}\right)(50.0\ \text{m/s})$$

$$v_{1f} = -0.025\ 0\ \text{m/s}$$



Figure 6.9 (Example 6.3) An archer fires an arrow horizontally to the right. Because he is standing on frictionless ice, he will begin to slide to the left across the ice.

■ SUMMARY

6.1 Momentum and Impulse

The **linear momentum** $\vec{p}$ of an object of mass m moving with velocity $\vec{v}$ is defined as

$$\vec{p} \equiv m\vec{v} \quad [6.1]$$

Momentum carries units of $\text{kg} \cdot \text{m/s}$. The **impulse** $\vec{I}$ of a constant force $\vec{F}$ delivered to an object is equal to the product of the force and the time interval during which the force acts:

$$\vec{I} \equiv \vec{F}\Delta t \quad [6.4]$$

These two concepts are unified in the **impulse–momentum theorem**, which states that the impulse of a constant force delivered to an object is equal to the change in momentum of the object:

$$\vec{I} = \vec{F}\Delta t = \Delta\vec{p} \equiv m\vec{v}_f - m\vec{v}_i \quad [6.5]$$

Solving problems with this theorem often involves estimating speeds or contact times (or both), leading to an average force.

6.2 Conservation of Momentum

When no net external force acts on an isolated system, the total momentum of the system is constant. This principle is called **conservation of momentum**. In particular, if the isolated system consists of two objects undergoing a collision, the total momentum of the system is the same before and after the collision. Conservation of momentum can be written mathematically for this case as

$$m_1\vec{v}_{1i} + m_2\vec{v}_{2i} = m_1\vec{v}_{1f} + m_2\vec{v}_{2f} \quad [6.7]$$



In an isolated system of two objects undergoing a collision, the total momentum of the system remains constant.

Collision and recoil problems typically require finding unknown velocities in one or two dimensions. Each vector component gives an equation, and the resulting equations are solved simultaneously.